COMMODITY ANALYST

How China Is Buffering the Global Energy Supply Shock

- We analyze Chinese energy balances since the start of the Middle East conflict and identify three reasons why a large hit to China's oil and natural gas imports—a key stabilizing force in LNG and especially oil markets during the Hormuz shock—led to only a moderate slowdown in overall economic activity (real Q2 GDP growth slowed to 3.6%, quarterly annualized, down from 5.3% in Q1, reflecting a negative fiscal impulse, higher energy prices, and adverse weather) with total energy demand growth remaining slightly positive at +0.4% YoY in April and May[1]:
- **First**, we observe **effective destocking of coal, oil, and natural gas**. The YoY growth in the effective use of coal/oil/natural gas stocks—especially slower coal restocking and faster implied oil destocking than this time last year—contributed +3.0/+2.2/+0.2 percentage points to total energy demand YoY growth in April and May.
- **Second**, **fuel substitution to coal and renewables** and the use of **power** has limited wider energy demand destruction. In April and May, China's lower use of oil/natural gas contributed -1.6/-0.1 percentage points to total energy demand YoY growth while greater use of coal/renewables contributed +1.4/+0.8pp. China's high EV penetration has allowed some switching to driving on electricity rather than gasoline: gasoline consumption was 23% lower and EV charging volume 60% higher YoY in April and May, buffering a larger fall in traffic.
- **Third, energy-related reductions in output have been concentrated in oil- and natural gas-intensive sectors**. Sectors less able to substitute the use of oil or natural gas have had to reduce physical output, while sectors with flexible underlying energy sources have shifted to or continued using alternatives such as power.

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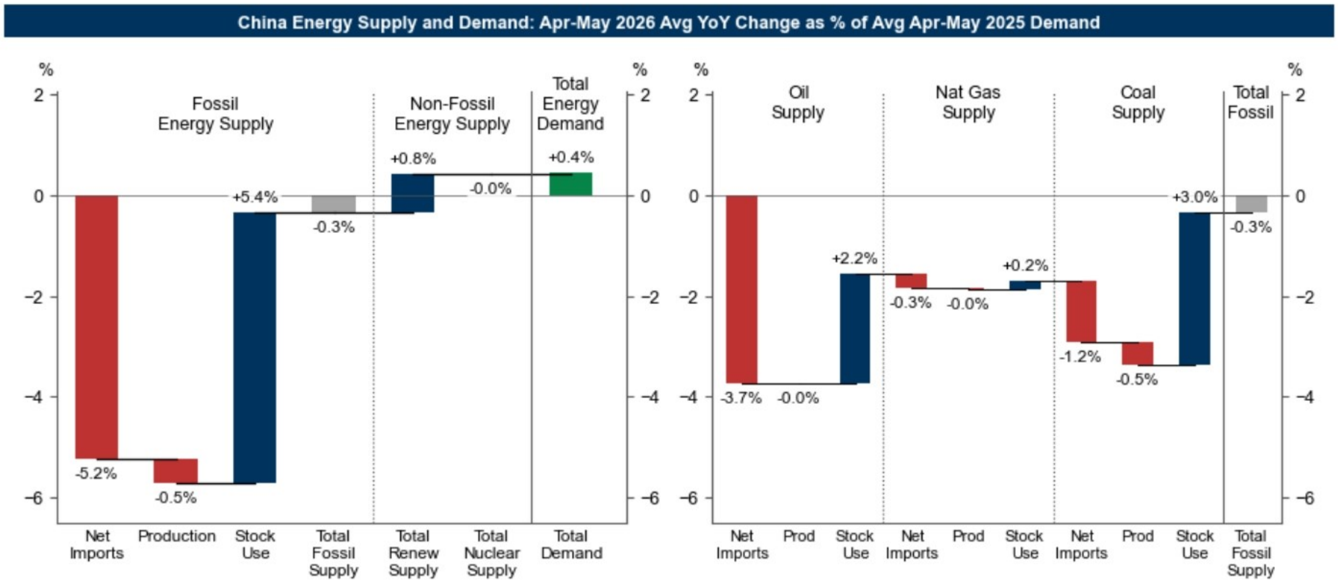
[1] "April and May" YoY growth refers to the growth in the April-May average total energy demand from 2025 to 2026. Percentage point contributions of each energy source below are calculated as the April-May average YoY energy demand or supply change for each fuel source, divided by the April-May 2025 average total energy demand.

Exhibit 1: Coal and Oil Effective Destocking and Higher Renewables Usage Helped Buffer China's Negative Fossil Fuel Import Shock

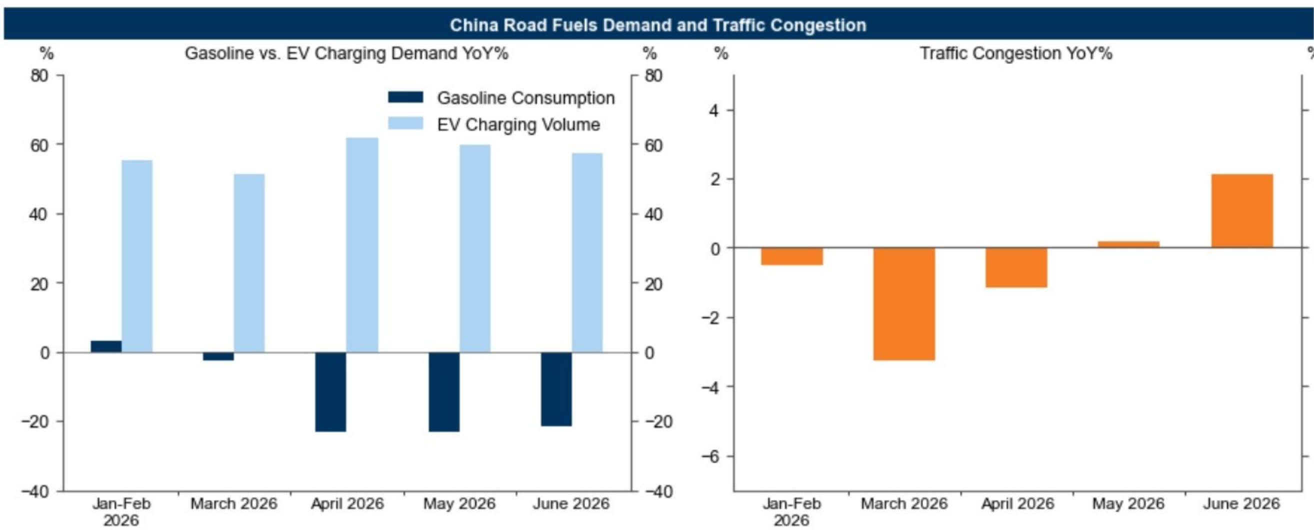
The units here are percentage point contributions to total energy demand growth.



"Oil Supply" is decomposed using changes in crude production, net imports, and inventory use, but scaled so that monthly total oil demand is aligned with monthly refined-products demand to better capture final energy consumption. "Stock Use" is the difference in average inventory drawdowns in April-May 2026 compared to April-May 2025, expressed as a percentage of average April-May 2025 total energy demand. We cite thermal coal inventory data from SX Coal. We cite SIA estimates of natural gas storage injections based on visible supply and demand. Owing to less visible oil inventories, we estimate oil inventory use as implied by NBS crude throughput and S&P products demand data, net imports data from Kpler and PetroLogistics, and production data from the IEA.

Source: IEA, Wind, SX Coal, SIA, Kpler, PetroLogistics, National Bureau of Statistics of China, Haver Analytics, S&P Global Market Intelligence, Goldman Sachs Global Investment Research

Exhibit 2: Traffic Congestion Remained Stable as Drivers Substituted Gasoline for Electricity



This references the Baidu map traffic congestion index which is calculated as the ratio of actual travel time to 'free flow' travel time, covering 98 cities.

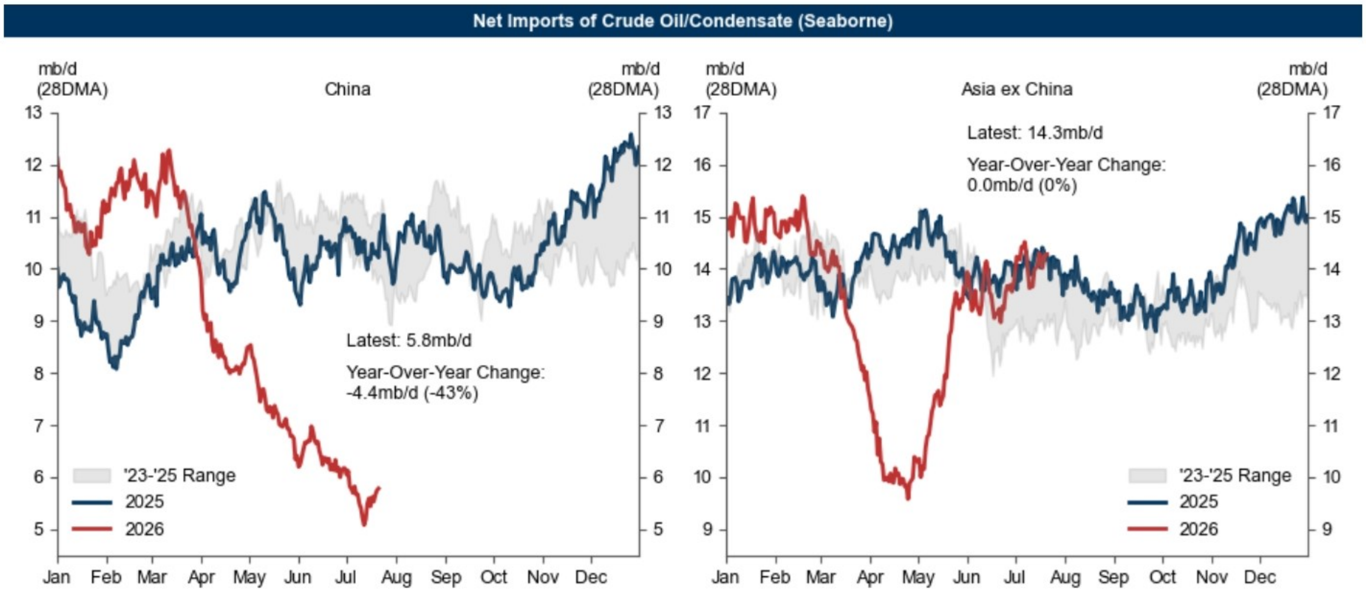
Source: Wind, Haver Analytics, Baidu, National Bureau of Statistics of China, National Energy Administration of China, Goldman Sachs Global Investment Research

How China Is Buffering the Global Energy Supply Shock

Major Fall in Net Imports, but Total Energy Demand Growth Still Positive
Ordinarily the largest importer of energy products shipped through the Strait of Hormuz, China has drastically reduced its net imports of fossil fuels, effectively acting as a shock absorber for global energy prices through reduced demand. Net imports of crude oil cratered in China and the rest of Asia beginning in March, but recovered in the rest of Asia to 2025 levels by June while continuing to fall in China through the first half of July (Exhibit 3).

China's net imports of oil/natural gas/coal fell 24%/7%/24% YoY in April and May reflecting YoY price jumps of 59%/49%/38% (Exhibit 4). These reductions in fossil fuel net imports were the largest source of negative total energy demand growth, representing -3.7pp/-0.3pp/-1.2pp of China's total YoY energy demand growth of +0.4% (Exhibit 5).

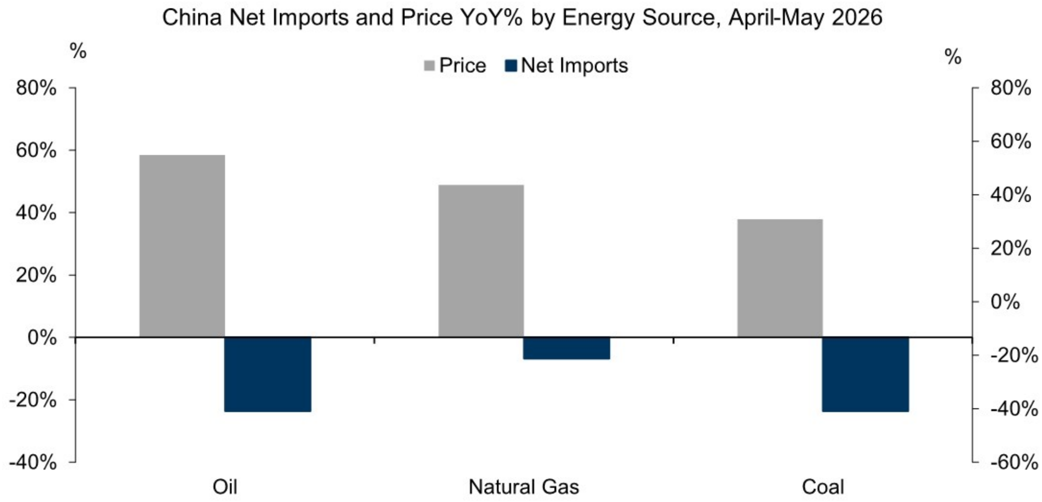
Exhibit 3: China Crude Oil Net Imports Continue to Fall While the Rest of Asia Recovers to 2025 Levels



Charts reference visible net imports of crude oil by sea from Kpler data.

Source: Kpler, Goldman Sachs Global Investment Research

Exhibit 4: China Fossil Fuel Net Imports Have Fallen Amid Higher Prices
YoY% Change in April-May Average, 2026



"Oil" net imports cites data on net seaborne crude imports from Kpler and pipeline crude imports from Petrologistics.

Source: Wind, SX Coal, SIA, Kpler, PetroLogistics, Platts, Goldman Sachs Global Investment Research

Chinese total energy consumption in April and May increased by an average of 0.4%, or 52 petajoules, year-over-year. To roughly estimate the impact of the supply shock on energy consumption, we estimate counterfactual consumption growth as the average +3.1% annual total energy demand growth rate from 2014-2023.[2] Applying this rate to China's average total monthly consumption in April and May 2025 would imply 375 PJ counterfactual YoY energy demand growth. This would suggest roughly 323 PJ of demand destruction for April and May, or 2.7pp reduction in the potential YoY growth rate. China's Q2 real GDP growth fell to 3.6% after 5.3% Q1 growth quarterly annualized, slightly exceeding our China team's nudged-down June forecast of 3.5% Q2 growth but missing market expectations. Lower GDP growth reflected mostly slower government spending, but also higher energy prices and unfavorable weather conditions.

Below, we highlight three factors that helped mitigate the total demand shock.

#1 Effective Destocking of Coal, Oil and Natural Gas Filled in for Fall in Fossil Fuel Imports and Production

Importing less of its energy needs from abroad, China has turned to its domestic inventories—rather than domestic production growth—to supplement the supply of fossil fuels.

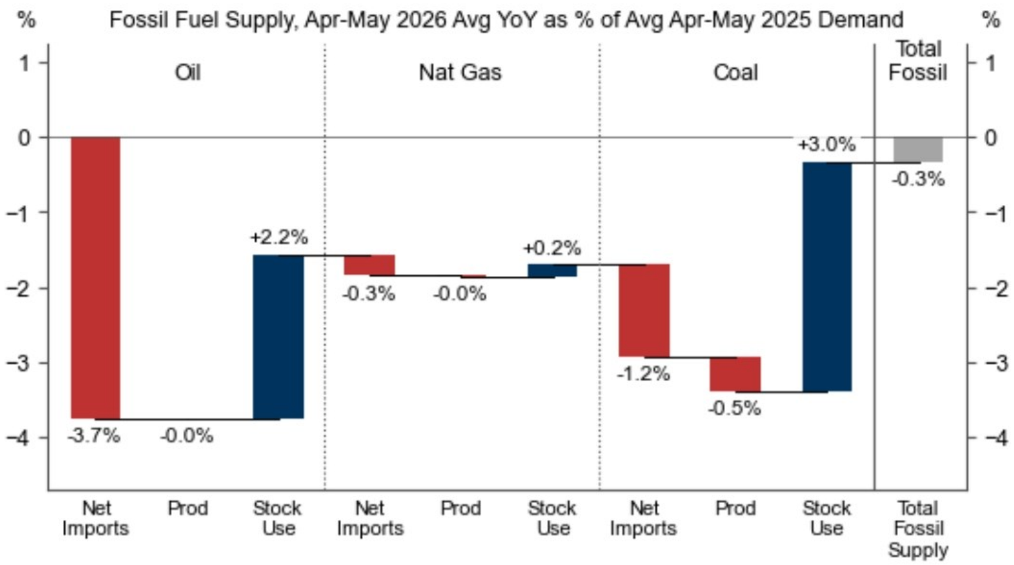
- Total domestic fossil fuel production actually fell slightly YoY in April and May, with lower coal production comprising a 0.5 percentage point reduction in total energy supply growth (Exhibit 5). Domestic crude oil production was unchanged compared

[2] YoY energy consumption growth rates vary by year and the month being analyzed. The 3.1% annual average growth rate for 2014-2023 looks at end-of-year balances. Though China's GDP growth has slowed somewhat in more recent years, we still think this average is a good benchmark for counterfactual April-May YoY energy demand growth, as it closely resembles the 3.7% YoY energy demand growth in January-February 2026 (just before the war) and the 3.2% YoY growth in April-May 2025.

to April and May of last year, likely constrained by high extraction costs in China's aging brownfields.

- The bulk of the rise in total energy consumption has been driven by the effective destocking of fossil fuels.
  - Thermal coal inventory levels increased by 1.6%/3.7% during April/May 2026, significantly lower than the 4.7%/5.8% MoM increase of April/May 2025. Though China's coal inventory level rose this April and May, we consider the reduction in MoM additions compared to last year's flows—in other words, how much less China added to its coal inventory this April/May compared to April/May 2025—as *effective* destocking. Defined this way, coal stock use contributed 3.0 percentage points to total YoY demand growth (Exhibit 5).
  - We estimate that oil destocking also accelerated, contributing 2.2pp to total YoY demand growth (Exhibit 5). Moreover, changes in China's *visible* crude oil stocks also appear directionally consistent with our implied destocking estimates of around 1mb/d in May and June, suggesting a shift from restocking in Q2 2025 to greater inventory use this year (Exhibit 6).
  - Effective natural gas destocking accounted for 0.2pp of total YoY energy demand growth (Exhibit 5).

Exhibit 5: Effective Fossil Fuel Destocking Filled in for Lower Imports and Production
The units here are percentage point contributions to total energy demand growth.



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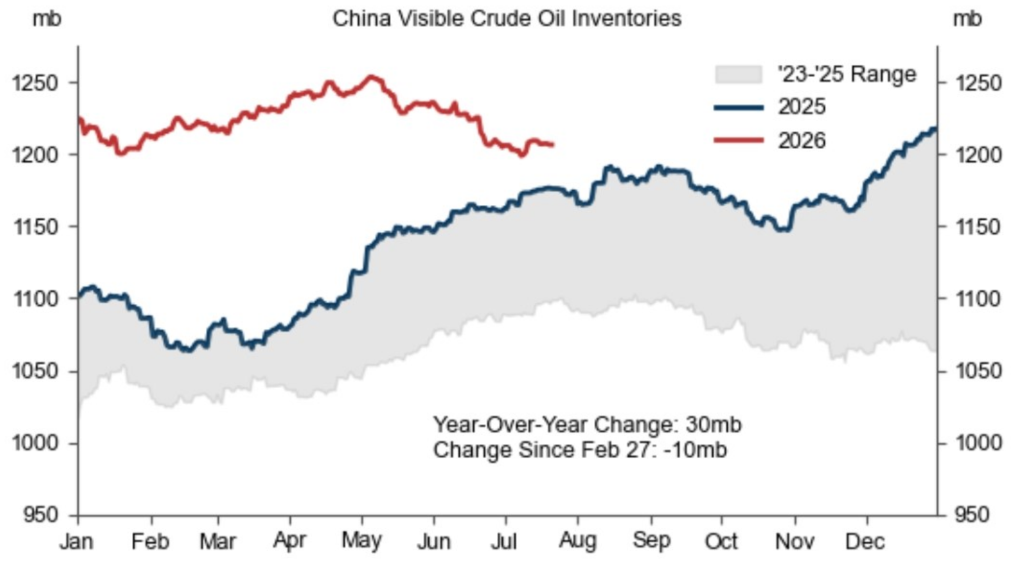
Source: IEA, Wind, SX Coal, SIA, Kpler, PetroLogistics, National Bureau of Statistics of China, Haver Analytics, S&P Global Market Intelligence, Goldman Sachs Global Investment Research

Exhibit 6: China's Visible Crude Inventories Grew in 2025 but have Fallen During the War



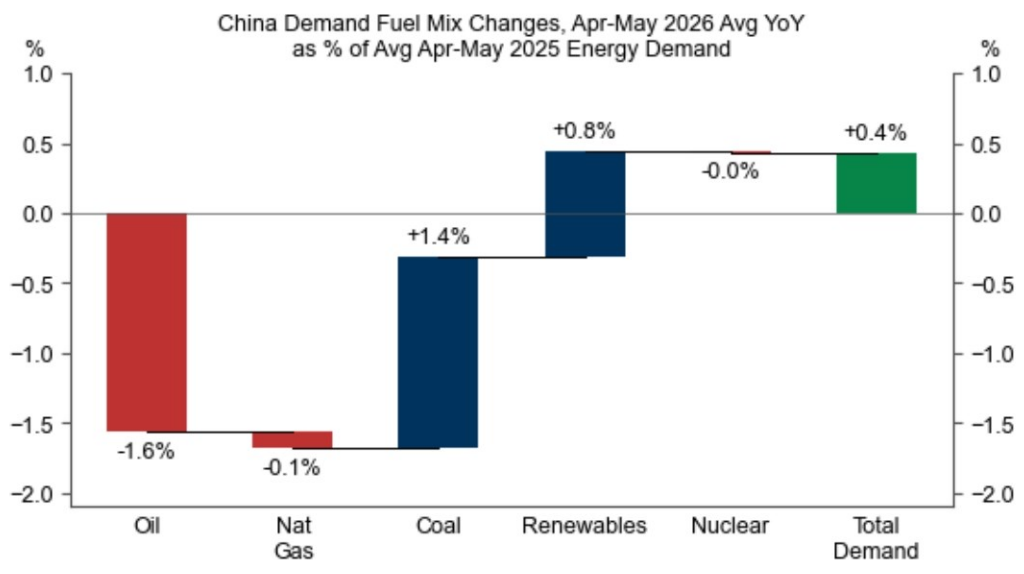
Source: Kpler, Goldman Sachs Global Investment Research

#2 Fuel Substitution to Coal and Renewables Has Limited the Demand Destruction

To avoid wider demand destruction caused by lower fossil fuel imports and production, China has increased its reliance on coal and renewables in its wider energy mix. Lower oil/natural gas use in China's overall energy demand contributed -1.6/-0.1 percentage points to its total YoY energy demand growth in April and May, while greater reliance on coal/renewables contributed +1.4/+0.8pp (Exhibit 7).

Exhibit 7: Coal and Renewables Pick Up the Slack of Lower Oil Supply

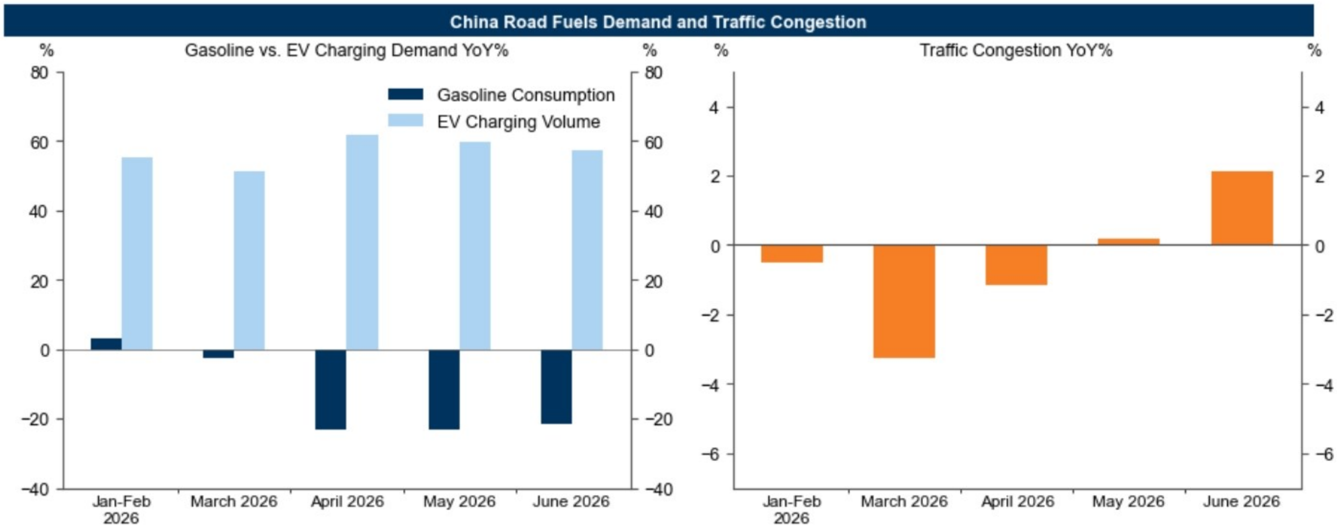
The units here are percentage point contributions to total energy demand growth.



Source: IEA, Wind, SX Coal, SIA, Kpler, National Bureau of Statistics of China, Haver Analytics, S&P Global Market Intelligence, Goldman Sachs Global Investment Research

As an example of this fuel switching in practice, we observe China substituting driving with gasoline for driving with electricity. Gasoline consumption fell 23%/23%/21% YoY in April/May/June, but EV charging growth rose to 62%/60%/57% YoY. Despite much lower gasoline consumption, traffic congestion remained relatively stable, falling only 1.2% YoY in April before growing by 0.2% and 2.1% YoY in May and June (Exhibit 8). These findings are consistent with our prior reporting on China's uptick in domestic EV sales since the start of the Iran war (despite seasonally-adjusted total passenger car sales remaining flat) and may reflect substitution both in car purchases (more EVs bought) and especially in choosing which kind of energy to drive on.[3]

Exhibit 8: Large Fall in Gasoline Consumption, but More EV Charging Keeps Traffic Relatively Stable



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Source: Wind, Haver Analytics, Baidu, National Bureau of Statistics of China, National Energy Administration of China, Goldman Sachs Global Investment Research

#3 Energy-Related Reductions in Output Are Concentrated in Oil- and Natural Gas-Reliant Sectors

Several industries that are highly oil- or natural gas-intensive have slowed production. Physical output of processed crude oil fell by 10.9% YoY in Q2 reflecting lower crude oil inputs (Exhibit 9).[4] Sulfuric acid, produced as a byproduct during oil and natural gas refining, saw 4.6% lower Q2 physical output YoY. Chemical fibers, produced with either oil or natural gas feedstocks like ethane or naphtha as inputs, saw 3.7% lower Q2 physical output YoY.

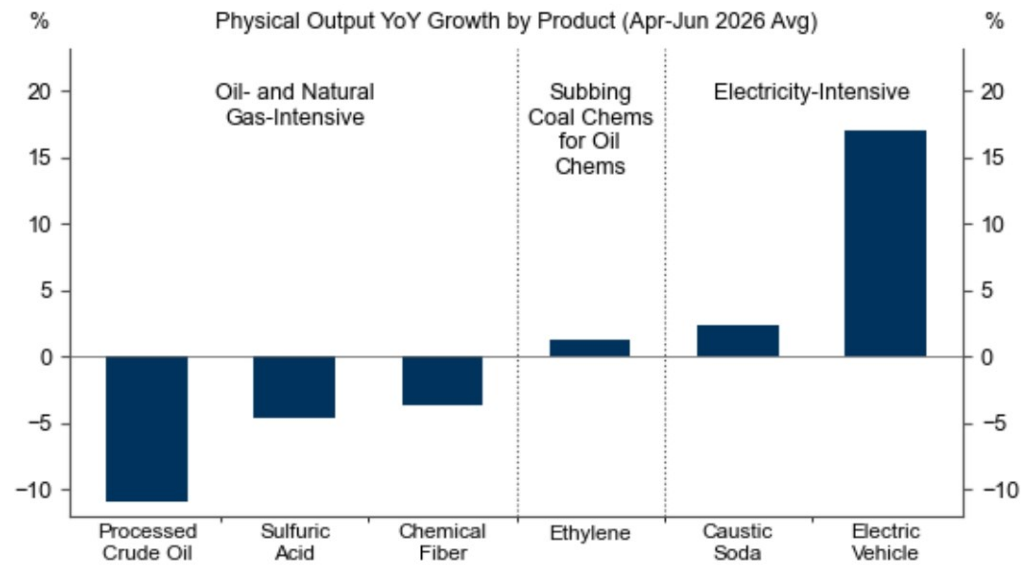
The production of the industrial chemical ethylene increased in Q2 by 1.2% YoY, rebounding from a 4.1% YoY fall in April to +2.1% and +5.5% YoY growth in May and June. Though conventional ethylene production involves steam cracking of oil feedstocks like ethane or naphtha, the recent rebound in ethylene output growth may reflect China's significant acceleration in modern coal-to-chemicals pathways like Coal-to-Olefins (CTO) where coal is gasified into syngas, synthesized into methanol, and dehydrated to form ethylene. China's use of coal in chemical production rose by 11.5% in April YoY amid the energy supply shock according to DBX Commodities, with coal-to-chemicals facilities residing atop domestic coal reserves well-positioned to facilitate the transition.

[3] For example, owners of Plug-in Hybrid EV cars may choose to charge on electricity rather than drive on gasoline, and households that own both EV and ICE cars may opt to drive their EVs more frequently. Usage of electric ride hailing services, trucks, buses, and small boats may also have picked up.

[4] This section reports the averages of April YoY%, May YoY% and June YoY% change in physical output.

Furthermore, energy-intensive products more reliant on power than oil or natural gas feedstocks saw more resilient output growth. The production of caustic soda, a major industrial chemical, is highly electricity-intensive but does not require oil or natural gas as unique inputs. Physical output of caustic soda grew by 2.4% YoY in Q2. EV production, more reliant on power than on materials made with oil and natural gas, also increased 17.0% YoY.

Exhibit 9: Lower Physical Output for Oil- and Natural Gas-Reliant Sectors



Source: Wind, National Bureau of Statistics of China, Goldman Sachs Global Investment Research

The commodities research team would like to thank Milan Chander for his major contributions to this project during his internship at Goldman Sachs Global Investment Research in summer 2026.

Disclosure Appendix

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We, Hongcen Wei, Daan Struyven, Samantha Dart, Yulia Zhestkova Grigsby, Lavinia Forcellese, Laura Cyr, Alexandra Paulus, Xinquan Chen, Lisheng Wang and Chelsea Song, hereby certify that all of the views expressed in this report accurately reflect our personal views, which have not been influenced by considerations of the firm's business or client relationships.

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